

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459520.707 +/- 0.0054 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.212 +/- 0.042
Transit depth (Rp/Rs)^2	4.51 +/- 1.77 [%]
Orbital Inclination (inc)	83.18 +/- 2.0
Airmass coefficient 1 (a1)	0.925 +/- 0.02
Airmass coefficient 2 (a2)	0.061 +/- 0.03
Scatter in the residuals of the lightcurve fit is	2.17 %
Best Comparison Star	#2 - [610, 279]
Optimal Method	PSF photometry
Transit Duration (day)	0.057 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.212 ± 0.042 vs 0.1646 (deviation 1.13σ)
Duration fit vs published	0.057 d vs 0.0758 d (deviation 0.75σ)
Mid-time O–C	-12.2 min
Depth SNR	5.0
Residual scatter	2.17 %

Light curve

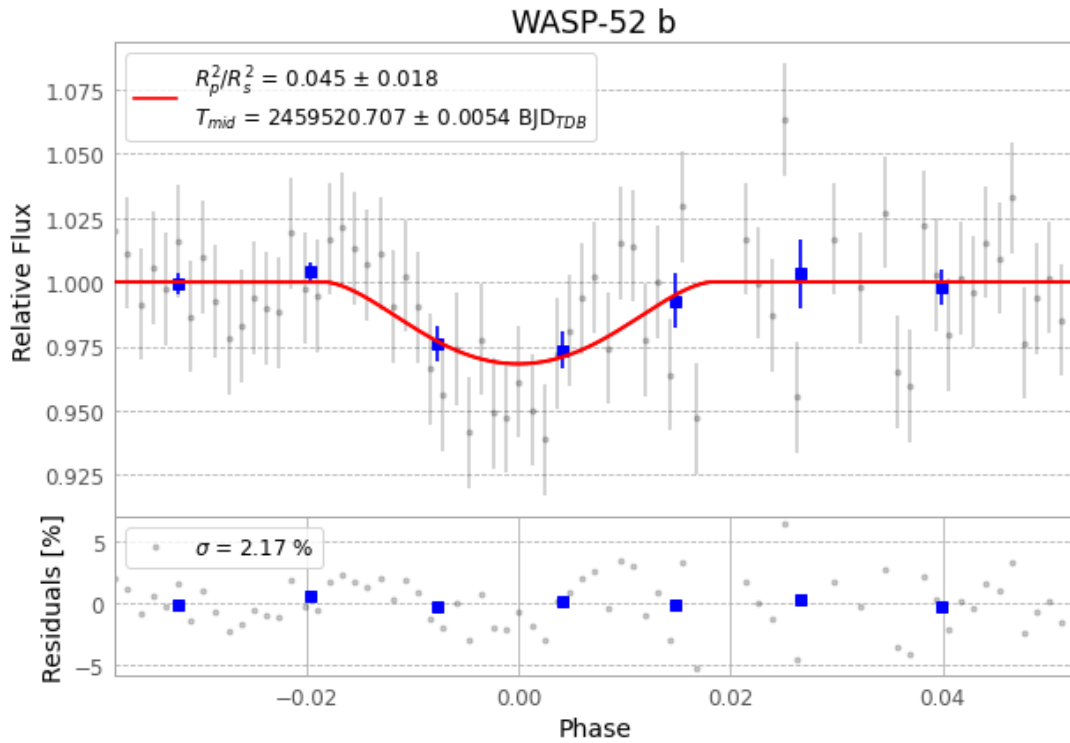


Figure 1 — FinalLightCurve_WASP-52 b_2021-11-01.png (SHA-256 eccb05df44ed64ec...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [492, 194], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8c599b952a59a9b8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52211102031512.FITS → WASP-52211102070312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-211102.log (4f068598eec9...), FinalLightCurve_WASP-52 b_2021-11-01.png (eccb05df44ed...), FinalLightCurve_WASP-52 b_2021-11-01.csv (a52a17557549...)

Compute resources

Wall time 265.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 10:50Z.